

Computer Network I

February 02, 2020

February 04, 25

*Kursus & Krose

* OSI - Open Systems Interconnection

→ Provided by ISO.

→ OSI Model has 7 layers.

i) Application

ii) Presentation

iii) Session

iv) Transport

v) Network

vi) Data link

vii) Physical

→ This model is the theoretical approach for data communication

* → TCP/IP is a protocol suit which is inspired by OSI.

→ It is the practical approach. The layers are as

i) Application

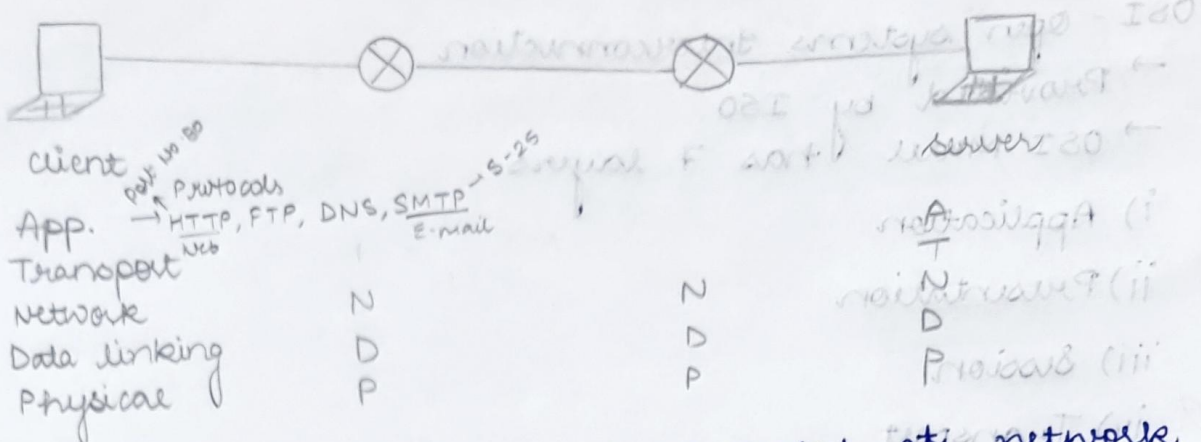
ii) Transport

iii) Network

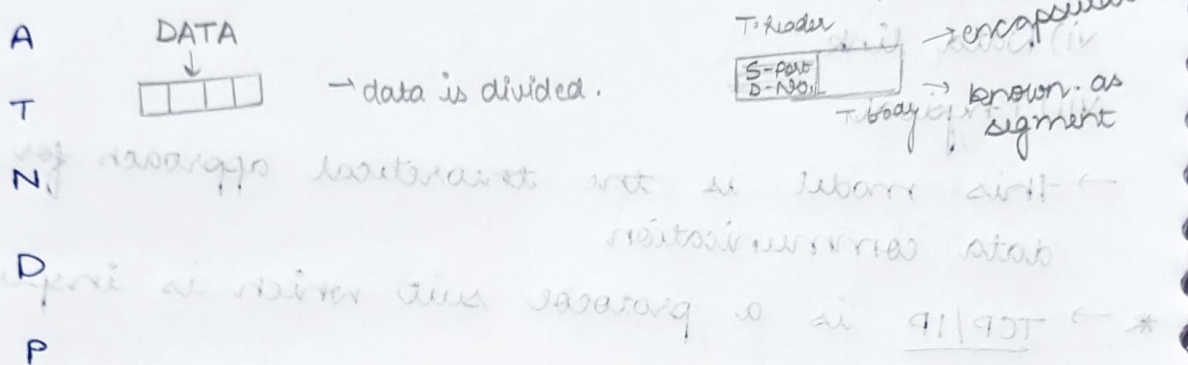
iv) Data link

v) Physical

→ Network architecture of Internet is the TCP/IP protocol suit.

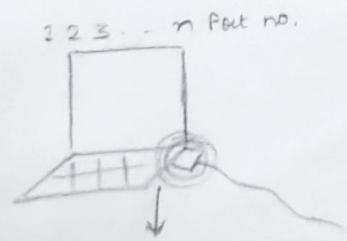


* The protocol suit will be implemented at network edges.



* The process of adding data

- # Transport layer
- Responsible for process to process deliveries
 - Port No. which is 16-bits long used to identify applications
 - Transportation → Reliable service → TCP
 - Unreliable service → UDP
 - Connection oriented service
 - Connection less service



→ Total Port nos. $2^{16} \rightarrow 0 \text{ to } 65535$

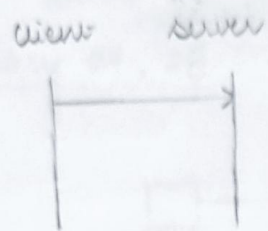
Servers → 0 - 1023 will known port no. reserved for servers.

1024 - 65535 → Eph

Here the data will not be transported to various processes using Port No. & this is called as process to process deliveries.

* UDP Protocol provides best effort but didn't provide data guarantee whereas TCP provides guaranteed delivery of data.

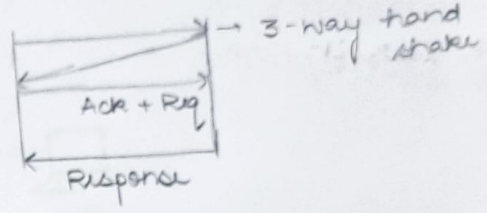
UDP



→ Can't divide data as it doesn't provide sequencing facilities.
DNS App.

TCP

Client Server



DNS: client

Requests - URL,
H. Name

K.O.B.

DNS: server

Response - IP

NIC → Network Interface Card

MAC → Medium Access Control

IP → 32 bit no. separated by 4 octates (each moves from 0-255).

→ network ID + host ID.

same ID same network.

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A
T
N
D
P

N
D
P

N Router
D Switch
P HUB

A

T

IP
MAC

Physical layer

→ App. & transport layers are not needed at network core because there is no application at this level & no process-to-process deliveries.

→ In one transmission no. of times each layer is used as, A → 2, T → 2, N → 4, D → 6.

Router

→ most intelligent / smart device.

→ used to connect two or more different networks.

→ store & forward. (different network IDs).

Switch

→ particularly used for creating LAN.

→ If no. of hosts is large, one switch can connect with different switches for scaling up.

→ It understands frames & uses store & forward device & has self-learning capacity.

→ Broadcast domain (stores the entire frame &

then broadcast to all ports.

→ Maintains self-learning table, which ~~switch~~ port is connected to which nodes / computer.

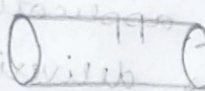
→ Default, learn then unicast.

HUB

- not a store & forward device.
- doesn't understand frames, only understands single bit of information.
- Creates LAN at the physical layer (level).
- Can't be used for large networks.

* switch

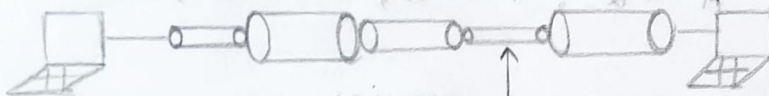
$$\text{Bandwidth} = \text{bits} / \text{sec}$$



cross sectional area,

no. of bits transferred per second

Throughput = Actual data transmission rate between the source & destination. (narrowest bandwidth)

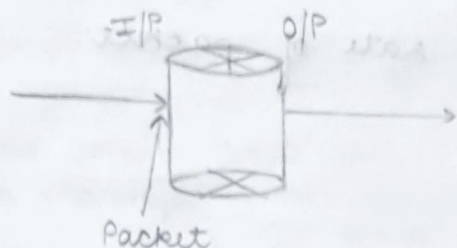


Throughput

End to end delay: (latency)

1. Processing delay
2. Queuing delay
3. Transmission delay
4. Propagation delay

* Processing delay



Router table

Destination IP	Serial Port

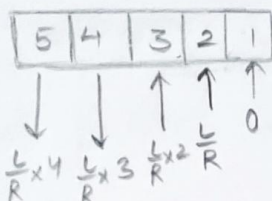
time required by the router to transfer from I/P to O/P port.

* Queuing delay

Buffer at the router.

time spent by a packet in queue.

Follows first come first serve.



Q-delay =

* Transmission delay:

Time required by a router to push all the bits of the packet on to the

$$d_T = \frac{L}{R}$$

where $L \rightarrow$ no. of bits present in a packet.

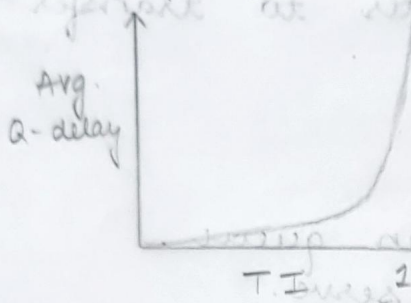
$R \rightarrow$ Transmission rate (bits/sec)

$$\left(\frac{2+0.5}{R} \right) \left(1 - \frac{1}{2} \right) + \frac{1}{2} \times \left(\frac{2+0.5}{R} \right) =$$

Traffic Intensity: (TI)

$\frac{\lambda_a}{R}$ $\lambda_a \rightarrow$ average arrival rate of packet.
 $\lambda_a \rightarrow$ arrival rate
 $R \rightarrow$ departure rate

$$\lambda_a < R \rightarrow \frac{\lambda_a}{R} \leq 1$$



When $TI > 1$, buffers start getting filled for all ports & the network starts dropping packets known as packet loss.

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* Propagation delay:

$$PD = \frac{\text{Distance (m)}}{\text{Speed (m/s)}}$$

$\rightarrow$ time required by single bit of information to reach from source to destination.

P33



File size = F bits R bps

Transmission Rate = R bps.

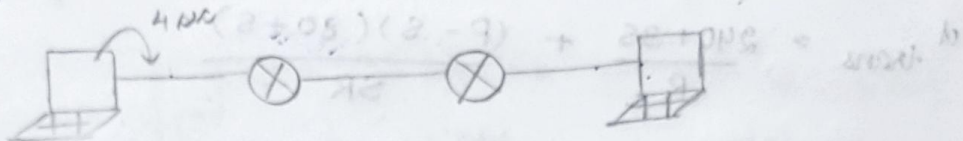
packet size = $80 + S$ bits

No. of packets = $\frac{F}{S}$

$$d_{trans} = \frac{80 + S}{R}$$

$$= \left(\frac{80 + S}{R} \right) \times 3 + \left(\frac{F}{S} - 1 \right) \left(\frac{80 + S}{R} \right)$$

P31



Packet size = 8×10^6 bits

link = 2 Mbps

$$a) d_{trans} = \frac{L}{R} = \frac{8 \times 10^6}{2 \times 10^6} = 4 \text{ sec.}$$

$$A \rightarrow B = 4 \times 3 = 12 \text{ sec}$$

b) 800 packets

1 packet = 10,000 bits

$$\rightarrow \frac{10^4}{2 \times 10^6} = 5 \text{ ms}$$

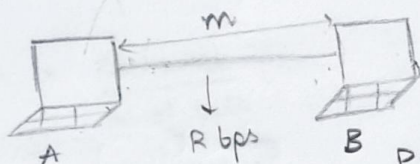
second packet = 10 ms

c) 15 ms + 799 x 5 ms

$$= 4010 \text{ ms}$$

$$= 4.01 \text{ sec.}$$

P6



P. size = L bits
Speed = S m/s

$$a) d_{prop} = \frac{m}{S}$$

$$b) d_{trans} = \frac{L}{R}$$

$$c) \text{Total delay} = \frac{m}{S} + \frac{L}{R}$$

d) either on link or at B.

e) at $t = d_{trans}$, and $d_{prop} > d_{trans}$, on the link.

f) $d_{prop} < d_{trans}$, first bit at B. because $\text{speed}_{prop} > \text{speed}_{trans}$.

$$g) S = 2.$$

$$d_{trans} = \frac{240+3S}{R} + \frac{(P-S)(80+S)}{SR}$$

$$= \frac{240+3S}{R} + \frac{80P + SF - 80S - S^2}{SR}$$

$$= \frac{240S + 3S^2 + 80P + SF - 80S - S^2}{SR}$$

SR

$$= \frac{160S + 2S^2 + 80P + SF}{SR}$$

SR, m 01 = taking bracket

$$= \frac{160}{R} + \frac{4S}{R} + \frac{F}{R} +$$

$$\frac{m}{2} = \text{gap to (a)}$$

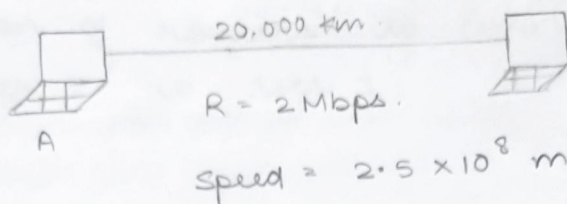
$$\frac{1}{R} = \text{gap to (b)}$$

$$\frac{1}{2} + \frac{m}{2} = \text{gap to (c)}$$

(a) either on link or at B
(b) at A = where, and to gap > 1 means, because gap > 1, then
(c) gap > 1 means, first at B, then to B
(d) 2 - 2

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P25



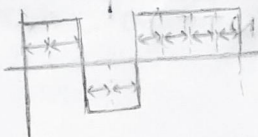
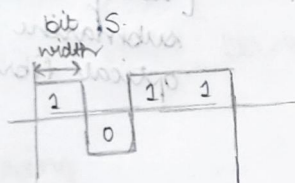
* $\text{Bandwidth} \times \text{delay product} = \text{Max no. of bits on a link at a given time.}$

a) $R \cdot d_{\text{prop}} = \frac{2 \times 10^6 \times 20,000 \times 10^3}{2.5 \times 10^8} = \frac{4 \times 10^6}{25} = 160 \text{ Kb}$

b)

c)

d)



source & destination should maintain the same bit width for correct data interpretation.

Bit width = $\frac{\text{Distance}}{\text{no. of bits}}$

$= \frac{20000 \times 10^3}{160 \times 10^3}$

$= 125 \text{ m} > \text{football field.}$

P29

Propagation

speed = $2.4 \times 10^8 \text{ m/s}$

$R = 10 \text{ Mbps}$

dist = 35786 km

a) $d_{\text{prop}} = \frac{35786 \times 10^3}{2.4 \times 10^8} = 0.1491$

b) $R \cdot d_{\text{prop}} = 10 \times 10^6 \times 0.1491 = 1.491 \text{ Mbits}$

c) $a_{trans} = 60 \text{ sec.} = \frac{L}{R}$

$L = 60 \times 10^7$

$L = 600 \text{ Mbits}$

INTERNET.

→ It's a network of networks.

ISP → Internet Service Provider

Access ISP (ISP1, ISP2, ISP3, ..., ISPn) --- Host 1, Host 2, Host 3, ..., Host n.

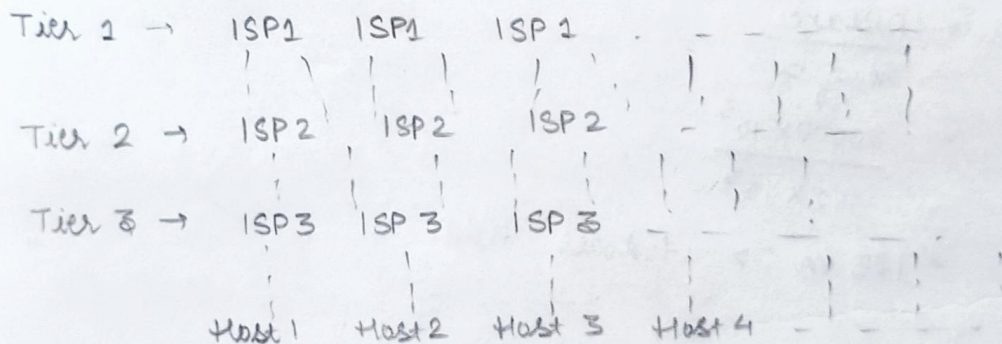
* Access ISPs are known as tier 3 ISPs.

* Hosts are connected to Tier-3 ISPs.

→ Tier 1 ISP → Global access providers.
(AT&T, Level 3, SPRINT, NTT)

[using satellites,
submarine cables,
optical fibre cables]

→ Tier 2 → Regional ISPs.



DNS → Distribution Network Systems.

→ CDN → Content Distribution Network

DASH → Dynamic Adaptive Streaming over HTTP.

* POP

- Point of presence.
- collection of routers to connect tier-3 to tier 2 and tier 2 to tier 1.

* Multi homing

- to provide redundancy.
- tier 3 ISP is connected to two or more tier 2 ISPs. If one lease line breaks, the connection & functionality will be preserved through another lease lines.

* IXP.

- Internet Exchange Point Nixi → India?
- Save the bandwidth and money.

* Peering

- Connection b/w same tier.
- All tier 1 ISPs are connected to each other.
↳ designed using mesh topology.

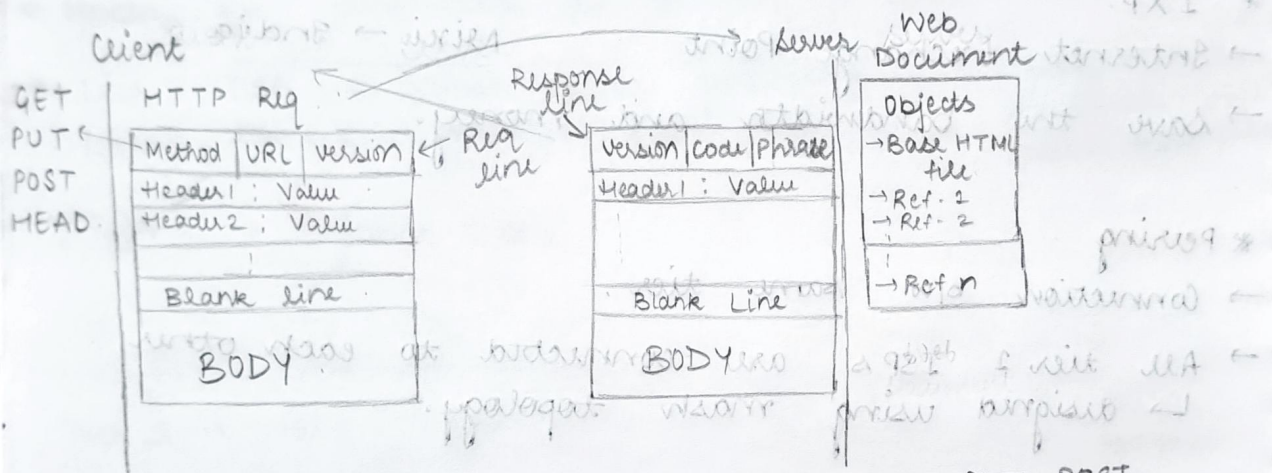
UNIT - 2

APPLICATION LAYER

→ E-mail, web, FTP, DNS, Bit torrent etc.

Web.

- Based on client server architecture.
- HTTP Port No. 80 Heart of web application
- HTTPS Port No. 443 Code at RFC 1945 & 2616 respectively.
- Uses TCP protocol ^{services} on transport layer.
- RFC < 1945 & RFC 2616.
↓
Request for comment.



* HTTP generally don't have body until we use PUT, POST.

* Client & Server communicate using headers.

* In Response line, code = 200 meaning successful
code = 404 meaning error.

Types of connections of HTTP :

i) Persistence

ii) Non-persistence

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HTTP Req Methods:

→ GET, POST, PUT, DELETE, TRACE, CONNECT, PATCH, OPTIONS

Request header Names:

→ AGENT

→ ACCEPT: Shows media format

→ ACCEPT-CHARSET:

→ ACCEPT-ENCODING:

→ ACCEPT-LANGUAGE:

→ ACCEPT-AUTHORIZATION:

→ HOST

→ DATE

→ UPGRADE

→ COOKIE

→ IF - MODIFIED - SINCE

Response Header Name :

→ DATE

→ UPGRADE

→ SERVER

→ COOKIE

→ CONTENT - ENCODING

→ CONTENT - LANGUAGE

→ CONTENT - LENGTH

→ CONTENT - TYPE

→ LOCATION

→ ACCEPT - RANGE

→ LAST - MODIFIED

→ COOKIE

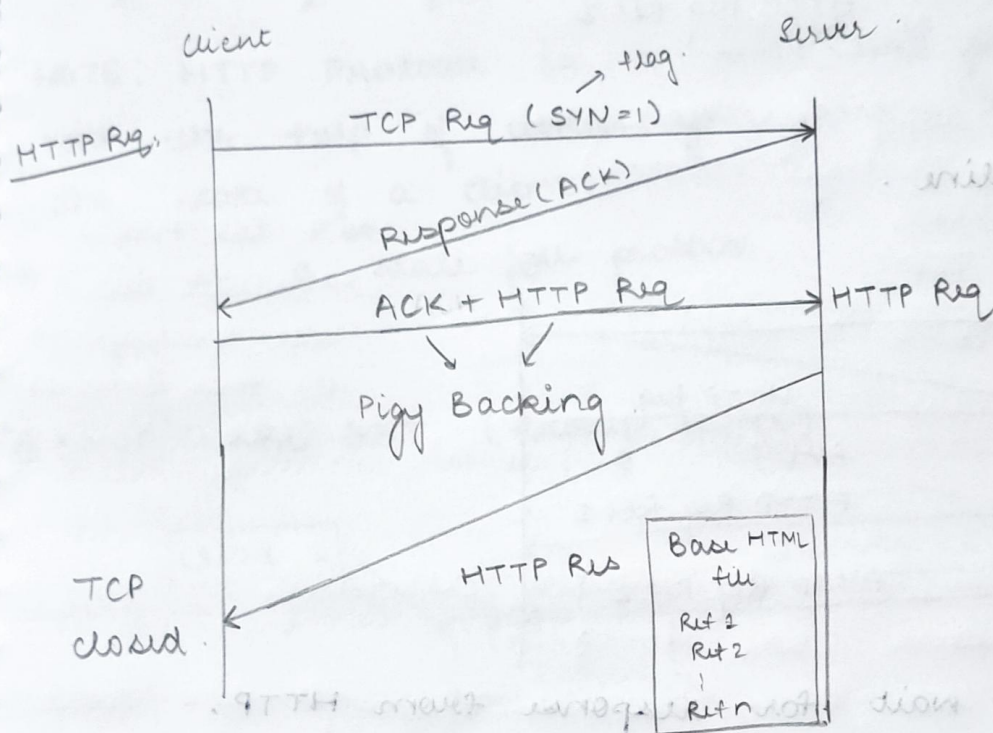
→ COOKIE AND SET COOKIE

→ CONDITIONAL

→ IF-MODIFIED - SINCE AND LAST-MODIFIED.

HTTP CONNECTION :

* Non - persistence

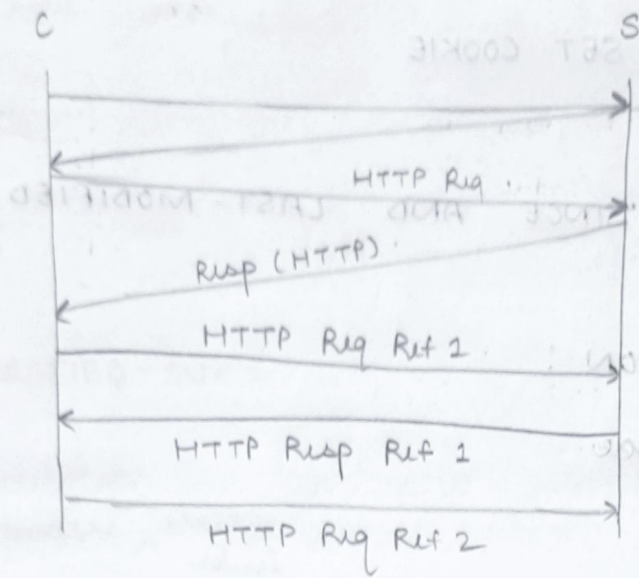


→ Ref 1 -- Ref n are the references to other objects in a document.

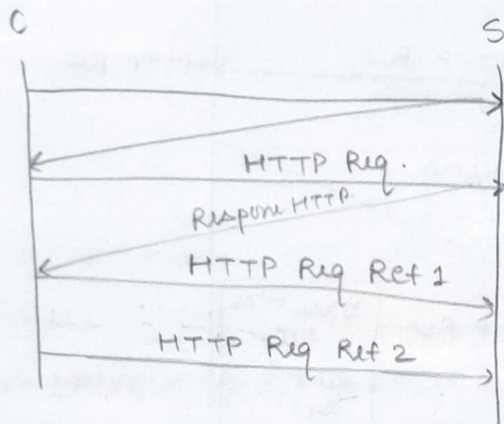
→ For every object, new TCP will be established.

* Persistence.

→ without pipeline.



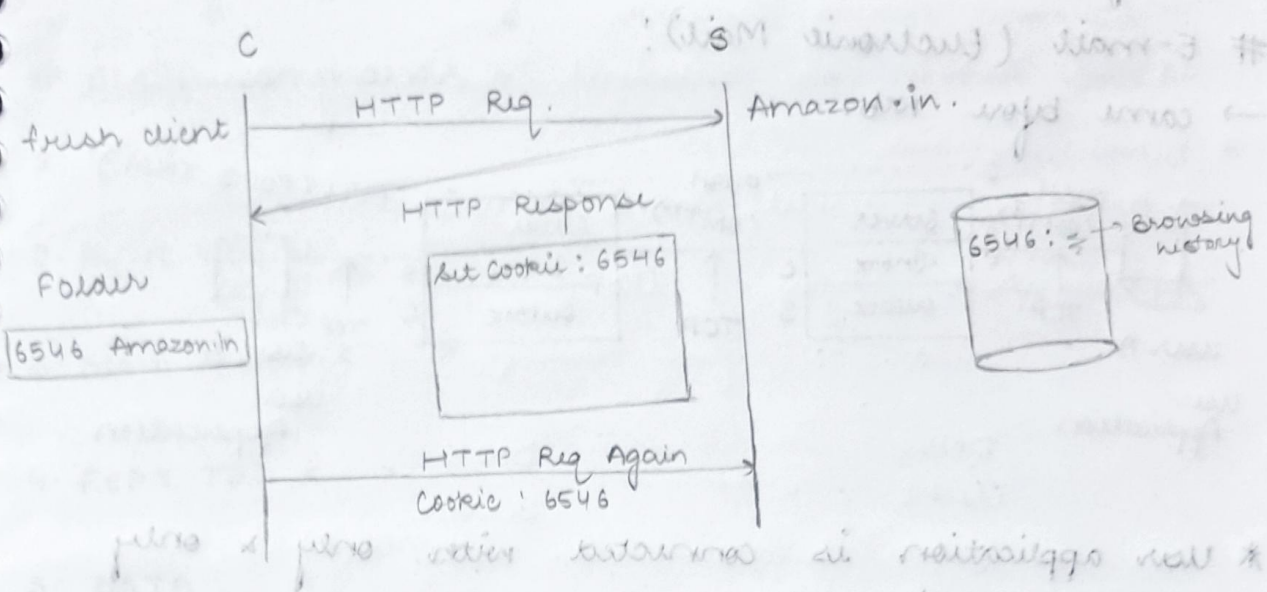
→ with pipeline.



* Doesn't wait for response from HTTP.

* DOS Attack.

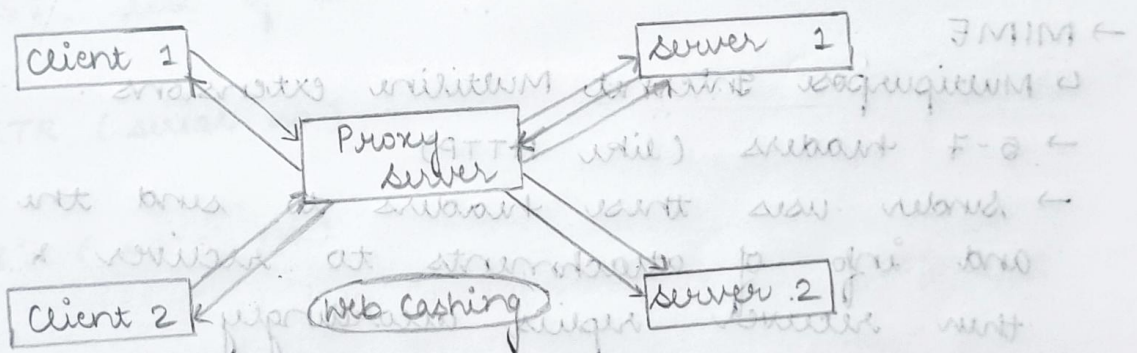
COOKIES. (Set Cookie & Cookie).



NOTE: HTTP Protocol is a state-less protocol. With the help of cookies, your server can maintain the state of a client.

* HTTP is a state-full protocol.

CONDITIONAL GET (Proxy Server)



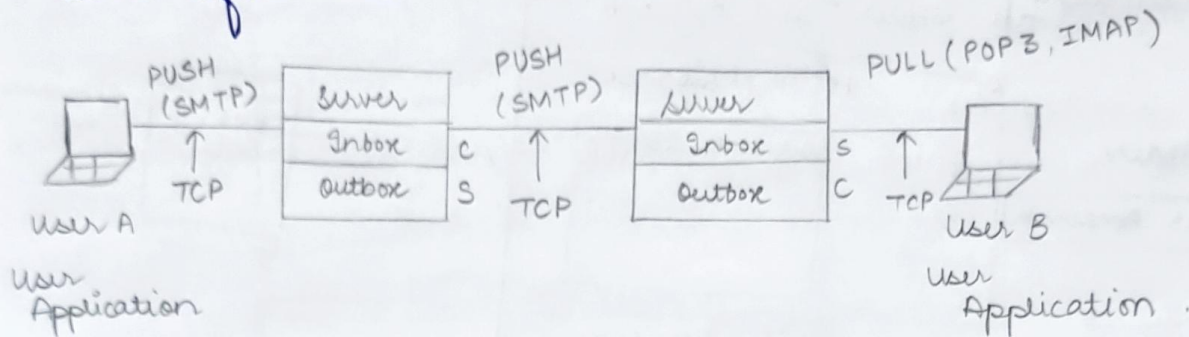
* Proxy server do web caching to keep a copy of request and response with it.

* Proxy server uses ~~if-modi~~ IF-MODIFIED-SINCE : Date & Time and ~~last~~ LAST-MODIFIED to check if the copy of document is fresh or stale.

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E-mail (Electronic Mail):

→ came before web.



* User application is connected with only & only 2 servers in between.

* SMTP is used for e-mail.
(Simple Mail Transfer Protocol)

* Client Server

* 7-Bit ASCII

Transport TCP

SMTP, POP3, IMAP

↓

Port No: 25, 110

→ earlier data was transferred in this format only.

→ MIME

↳ Multipurpose Internet Multiline extensions.

→ 5-7 headers (like HTTP)

→ Sender uses these headers to send the data and info of attachments to receiver & then receiver replies accordingly.

* SMTP command known as PUSH command.

* IMAP is latest version POP3 is older.

* From mail server to retrieve mail, we may use HTTP.

* But for PUSH, only SMTP is used.

SMTP commands

1. HELO

2. AUTH LOGIN

3. MAIL FROM: < >

4. RCPT TO: < >

5. DATA

6. .

7. QUIT

POP3/IMAP

1. LIST: list of all mails with serial nos.

2. RETR (serial no.)

3. DELE (serial no.)

4. QUIT

DNS.

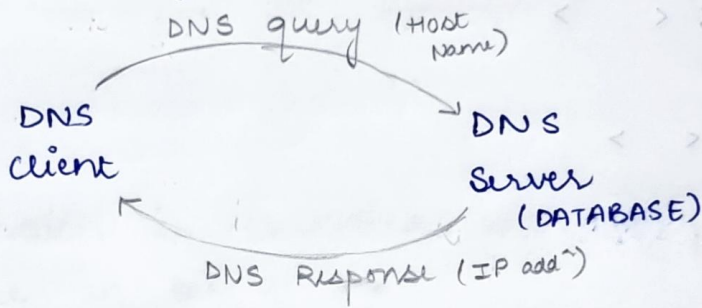
→ Domain Name System

→ Also known as Internet Directory services

* BIND Software

→ Berkeley Internet Name Domain.

→ Transport layer : UDP



→ Tasks performed by DNS :

1. Host Aliasing

→ Host name to IP address.

2. Mail server Aliasing

→ Mail server to IP address.

3. Load Distribution.

Amazon.in

Canonical Name (full Name)	IP add ⁿ of Amazon
	IP1
	IP2
	IP3
	Replica

→ Problems if DNS DB is centralized.

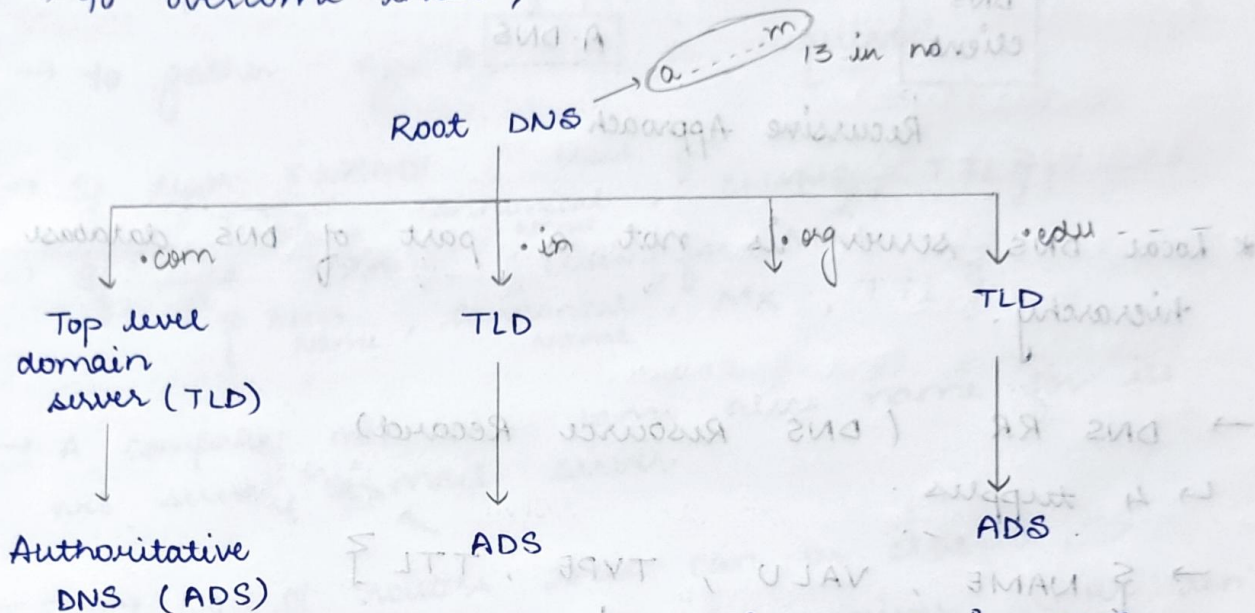
1. Single point of failure

2. Traffic

3. Distant location

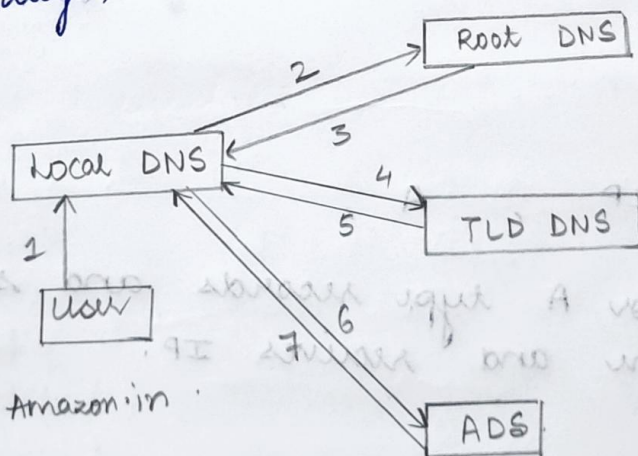
4. Maintenance issues

→ To overcome these, we use distributed DNS database.



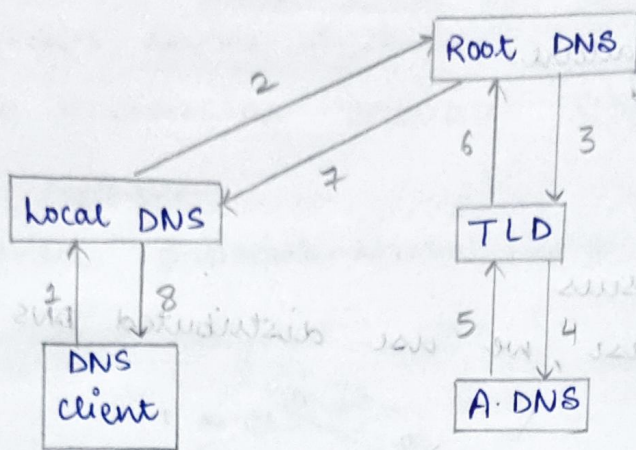
→ Most ROOT DNS servers are installed in DNS.
→ ADS have actual mapping of host name & IP.

query → Root DNS → Appropriate TLD server → Appropriate ADS.
↳ 2 ways, iterative & recursive.



* Local DNS works as a proxy server to for IP add^s caching.

Iterative Approach.



Recursive Approach

* Local DNS server is not a part of DNS database hierarchy.

→ DNS RR (DNS Resource Record)

↳ 4 tuples

→ { NAME, VALUE, TYPE, TTL }

↓
A
NS
CNAME
MX

Time to live

→ DNS RR → Messages that travel in the Database hierarchy.

* of type = A

{ Host Name, IP, A }

→ we looking for A type records and sends host name and receives IP.

→ If type NS

{ Domain Name, A DNS, NS, TTL }

VALU → Host name of an authoritative DNS server that knows how to obtain the IP add^r for host in the domain.

→ to gather Type A, NS is required.

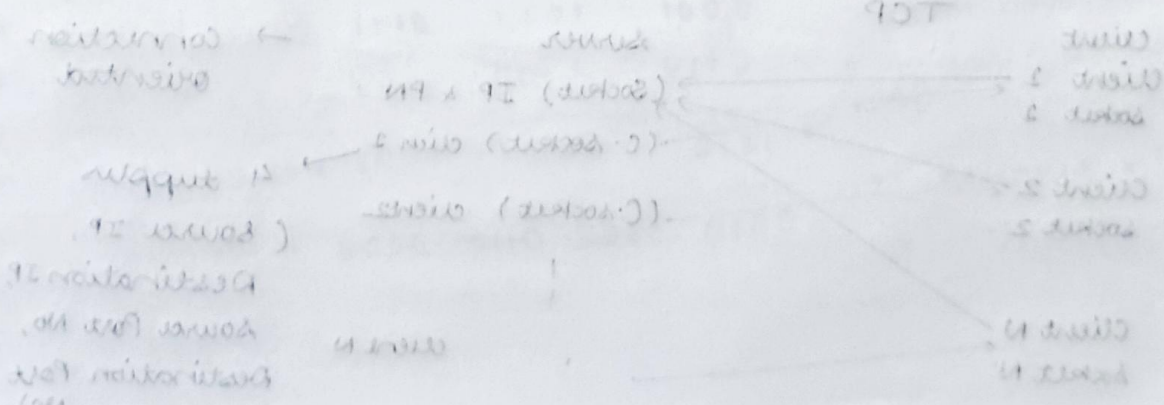
→ If type CNAME : Uses for web.
{ Alias Name, Canonical Name, CNAME, TTL }

→ If type MX : Uses for Mail
{ Alias Name, Canonical Name, MX, TTL }

→ A company can use same alias name for its web server & mail server.

TTL → No. of routers that can be crossed.

→ If for some reason if receiver server isn't found then data should not remain forever in the network. Therefore, in such cases TTL is useful.



TRANSPORT LAYER

TCP : Transmission Control Protocol

UDP : User Datagram Protocol

Application layer

Socket (API)

Transport layer

Socket : It is a gateway for a particular process.

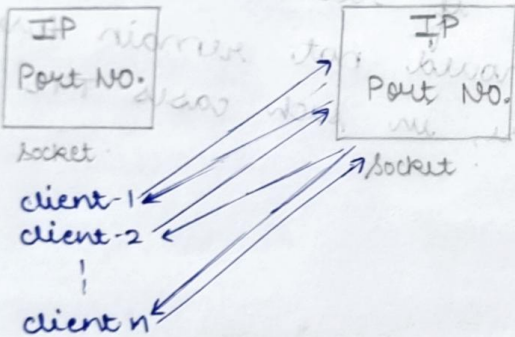
→ IP + Port Number.

→ UDP

Client

Server

→ Connectionless

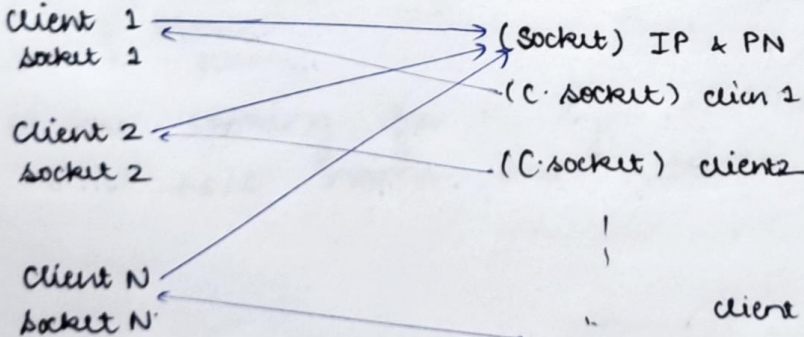


→ TCP

Client

Server

→ Connection oriented



4 tuples
(Source IP,
Destination IP,
Source Port No.,
Destination Port No.)

In TCP, a new single socket is created for each client.

UDP Header

header [	16	16	
	Source Port No.	Destination Port No.	32
	length	check sum	32
	Application Data (Payload)		

$$= 64 = 8 \text{ bytes}$$

→ check sum is 16 bits long & is used for detecting error in UDP segments.

To detect error through check sum:
* Divide data through in 16-bit word (may need to use padding).

* On sending side, add 16 bits source port no, destination port no, length & 16-bit word data. and set check sum as 1's complement of sum.

* On receiving side, add check sum as well and if all 16 bits are 1, this means no error.

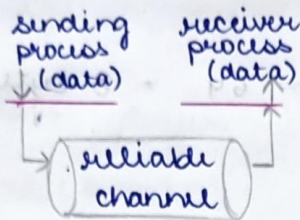
$$\begin{array}{r}
 \text{word}_1 = 1010 \ 0000 \ 1110 \ 1100 \\
 \text{word}_2 = 1010 \ 1110 \ 1101 \ 1000 \\
 \text{word}_3 = 1010 \ 1010 \ 0011 \ 0110 \\
 \hline
 1111 \ 1001 \ 1111 \ 1010 \leftarrow \text{wrap}
 \end{array}$$

$$1's \text{ complement} = 0000 \ 0110 \ 0000 \ 0100$$

+ Receiving = 1
No error.

Principles of reliable data transfer: (rdt)

application layer
transport layer



reliable data transfer protocol (sending side (packet))

rdt_send()

→ called from above, (eg. by app)

FSM: Finite State Machine

rdt1.0: reliable transfer over a reliable channel

wait for call from above

Sender

Receiver

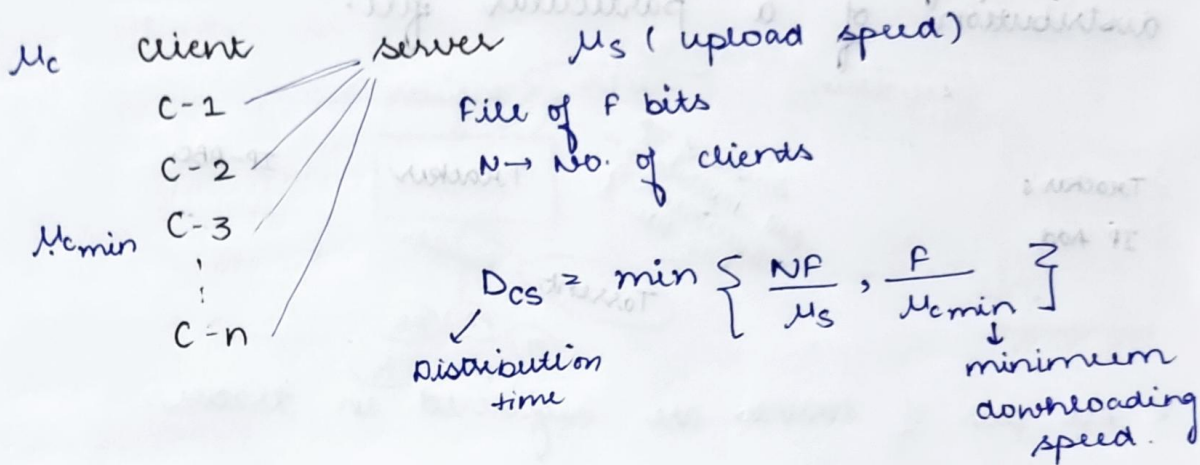
ARQ protocol type: Automatic Repeat Request

→ Cover this topic from PPT.

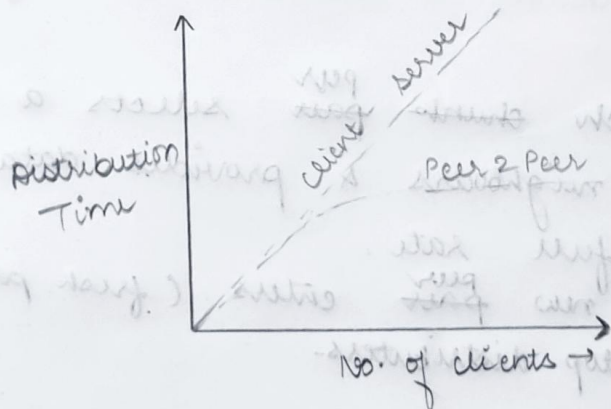
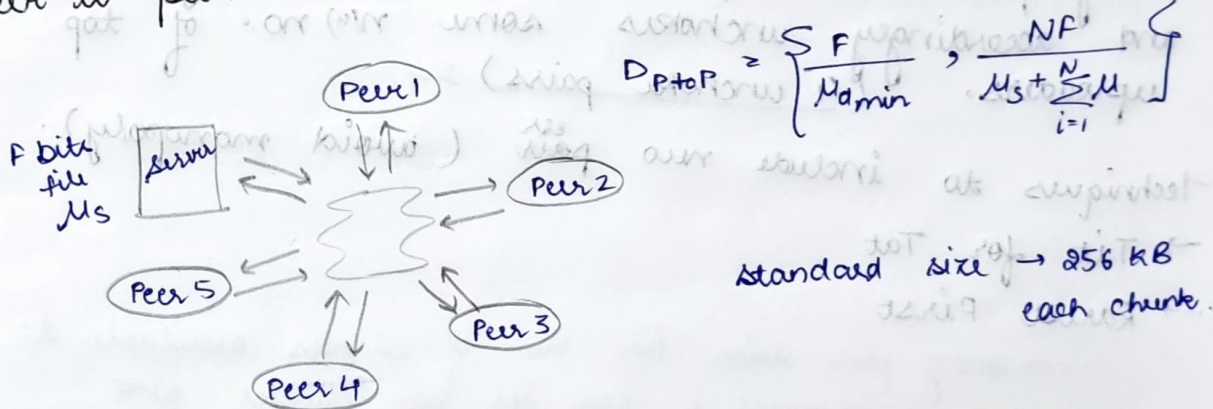
March 05, 25

BIT TORRENT

Client server



Peer to peer Architecture



Bit torrent:

Based on Peer 2 Peer architecture.

XYZ.torrent

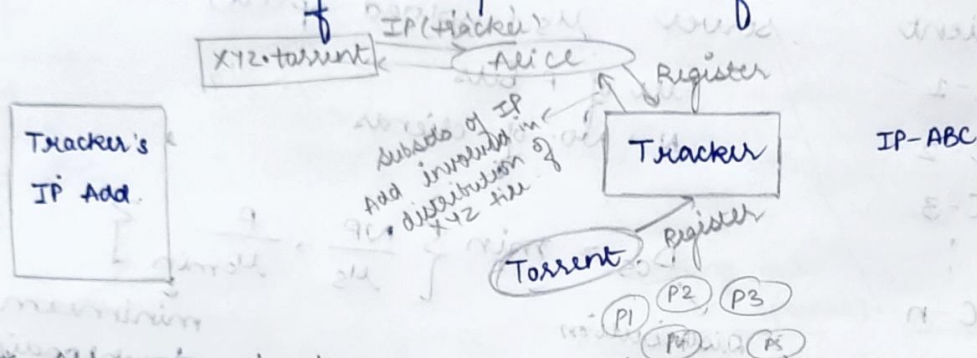
XYZ → movie to be downloaded

seeds:

leeches:

Torrent is

collection of peers which are involved in a distribution of a particular file.



* All pair of torrents are registered in tracker.

→ uses TCP as transportation services.

→ Every ~~peer~~ will keep a check on its neighbours and accordingly, unchokes some ~~no.~~ no. of top uploaders. (4 unchoke pairs)

Techniques to include new ~~peer~~ (avoid monopoly):

→ Tit for Tat

→ Rarest First

→ every pair asks for no. of chunks available to its neighbours & also its own chunks and identifies rarest chunks & distribute them first.

→ Every 10/30 seconds, each ~~chunks~~ ~~peer~~ selects a ~~optimistically unchoked peer~~ ~~random peer~~ from its neighbours & provides data to that ~~peer~~ will full rate.

Advantage: i) When a new ~~peer~~ enters (fresh pair).
ii) change in top distributors.

Choke peer → n-5